

Waveshaping & Distortion

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Introduction to Waveshaping

Yesterday we learned about filters.

Filters can *shape* a signal. They cannot add new frequencies.

Applying a nonlinearity *adds new frequencies to a signal*.

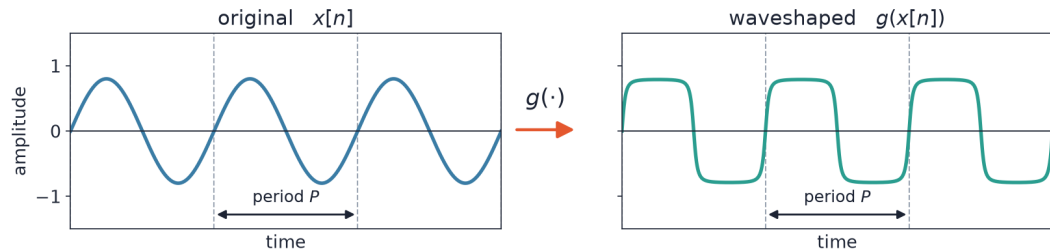
Introduction to Waveshaping

To add new frequencies to our signal, we have to feed it into a **nonlinear function**.

Suppose $x[n]$ is our digital signal, and $g(x[n]) = y[n]$ is our function that returns the output signal. Some common candidates are:

- ▶ **Hard clip:** $g(x) = \max(-1, \min(1, x))$
- ▶ **Cubic soft clip:** $g(x) = x - x^3/3$
- ▶ **Hyperbolic tangent:** $g(x) = \tanh(x)$
- ▶ **Algebraic sigmoid:** $g(x) = x/(\sqrt{1+x^2})$
- ▶ **Arctangent:** $g(x) = \frac{2}{\pi} \arctan(x)$
- ▶ **Sine wavefolder:** $g(x) = \sin(x)$

The Effect of Nonlinearities



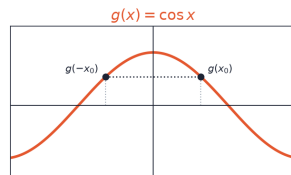
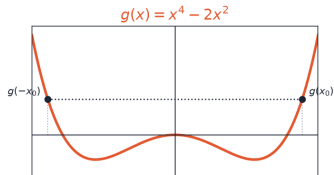
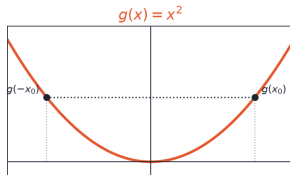
Suppose we apply waveshaping to some signal with period P . What happens?

Waveshaping functions have no memory. They cannot change the period of a signal.

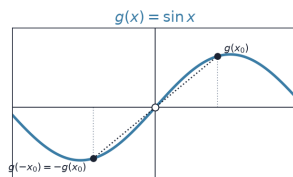
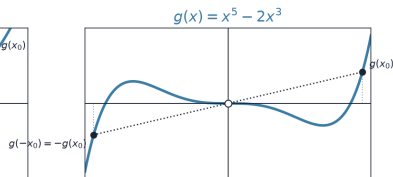
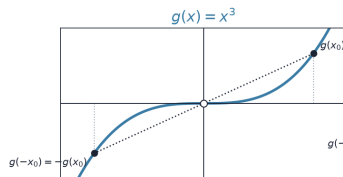
Therefore, the only frequencies that can be added are the *harmonics*. If $f_0 = 1/P$ the base frequency, then we'll only see $f_0, 2f_0, 3f_0, \dots$

Even vs Odd functions

EVEN: $g(-x) = g(x)$ — mirror across the y-axis



ODD: $g(-x) = -g(x)$ — 180° rotation about the origin



Odd vs. even

A function g is

$$\text{even} \iff g(-x) = g(x)$$

$$\text{odd} \iff g(-x) = -g(x)$$

Theorem: every function splits into an even part and an odd part. Define

$$g_e(x) = \frac{g(x) + g(-x)}{2}$$

$$g_o(x) = \frac{g(x) - g(-x)}{2}$$

Check the symmetries by substituting $-x$:

$$g_e(-x) = \frac{g(-x) + g(x)}{2} = g_e(x)$$

$$g_o(-x) = \frac{g(-x) - g(x)}{2} = -g_o(x)$$

and the two parts add back up to g :

$$g_e(x) + g_o(x) = \frac{g(x) + g(-x) + g(x) - g(-x)}{2} = g(x). \quad \blacksquare$$

Harmonics of even and odd functions

Suppose we feed a sinusoid with frequency f_0 into a distortion function $g(x)$.

It turns out that even distortion functions produce even harmonics at DC, $2f_0$, $4f_0 \dots$

Odd distortion functions produce f_0 , $3f_0$, $5f_0 \dots$

For example, suppose we feed in a 100 Hz sine wave:

- ▶ if $g(x)$ is **odd**, the output contains 100, 300, 500, 700 Hz, ...
- ▶ if $g(x)$ is **even**, the output contains DC, 200, 400, 600 Hz, ...

Weird!

Taylor Series

Any smooth function can be approximated, near $x = 0$, by a sum of monomial powers — its *Taylor series*:

$$g(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots \quad \text{where } a_n = \frac{g^{(n)}(0)}{n!}$$

We saw that even powers ($x^2, x^4, \dots$) are even functions, and odd powers ($x, x^3, \dots$) are odd. In fact:

$$g \text{ even} \implies g(x) = a_0 + a_2x^2 + a_4x^4 + \cdots \quad (\text{even powers only})$$

$$g \text{ odd} \implies g(x) = a_1x + a_3x^3 + a_5x^5 + \cdots \quad (\text{odd powers only})$$

Fun fact: a “sharper” distortion curve will need a longer power series to be approximated, which means more harmonics!

Powers of sinusoids

Now feed a sinusoid into the series: let $x = \sin \theta$ with $\theta = 2\pi f_0 t$. Then

$$g(\sin \theta) = a_1 \sin \theta + a_2 \sin^2 \theta + a_3 \sin^3 \theta + \dots$$

Trig identities reduce each power to plain sinusoids again:

$$\sin^2 \theta = \underbrace{\frac{1}{2}}_{\text{DC}} - \underbrace{\frac{1}{2} \cos 2\theta}_{2f_0}$$

$$\sin^3 \theta = \underbrace{\frac{3}{4} \sin \theta}_{f_0} - \underbrace{\frac{1}{4} \sin 3\theta}_{3f_0}$$

$$\sin^4 \theta = \underbrace{\frac{3}{8}}_{\text{DC}} - \underbrace{\frac{1}{2} \cos 2\theta}_{2f_0} + \underbrace{\frac{1}{8} \cos 4\theta}_{4f_0}$$

$$\sin^5 \theta = \underbrace{\frac{5}{8} \sin \theta}_{f_0} - \underbrace{\frac{5}{16} \sin 3\theta}_{3f_0} + \underbrace{\frac{1}{16} \sin 5\theta}_{5f_0}$$

In general $\sin^p \theta$ is a weighted sum of sinusoids at $p\theta$, $(p-2)\theta$, $(p-4)\theta$, ...

odd powers produce only **odd** multiples of f_0 , and **even** powers only **even** multiples (including DC).

Nonlinearities are nonlinear

Nonlinear functions are... nonlinear. Therefore, no superposition:

$$g(x_1 + x_2) \neq g(x_1) + g(x_2)$$

We can't analyze the effects of nonlinearities on real signal by considering one sinusoid at a time.

For example: feed two sinusoids into x^2 :

$$(\sin \theta_1 + \sin \theta_2)^2 = \sin^2 \theta_1 + \sin^2 \theta_2 + \underbrace{2 \sin \theta_1 \sin \theta_2}_{\text{cross term}}$$

$$2 \sin \theta_1 \sin \theta_2 = \underbrace{\cos(\theta_1 - \theta_2)}_{f_1 - f_2} - \underbrace{\cos(\theta_1 + \theta_2)}_{f_1 + f_2}$$

We see new components *sum and difference* frequencies (**intermodulation**). These are not clean multiples of any fundamental frequency.

More sinusoids in $\Rightarrow$ combinatorial explosion of cross cross terms $\Rightarrow$ dense inharmonic content.

Nyquist Wept

These new harmonics do not respect Nyquist. For example, if we hard-clip 5 kHz sine at $f_s = 48$ kHz, we get harmonics at

$$5, 15, \underbrace{25, 35, 45, \dots}_{\text{above } f_s/2 = 24 \text{ kHz}} \text{ kHz}$$

Do these vanish? No! They *alias* back to $f_s - f$:

$$25 \rightarrow 23 \text{ kHz} \quad 35 \rightarrow 13 \text{ kHz} \quad 45 \rightarrow 3 \text{ kHz}$$

They are very real and very audible.

Managing Aliasing

One way to deal with aliasing is **oversampling**, which gives us more room before we hit Nyquist.

1. **Upsample** to Nf_s : insert $N - 1$ zeros between samples, then low-pass at the *original* Nyquist $f_s/2$.
2. **Distort** at Nf_s
3. **Low-pass** at $f_s/2$ again, removing those harmonics that would otherwise alias when we downsample.
4. **Downsample**: keep every N th sample.

Not perfect; harmonics above the raised Nyquist $Nf_s/2$ can still fold.

Multi-band distortion lets us avoid distorting high frequencies, which tames the partials.

Interlude: splitting a signal in two

For multiband distortion we want to split the signal into a low-frequency band and high-frequency band such that

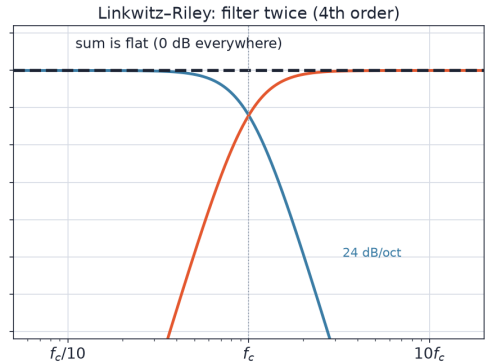
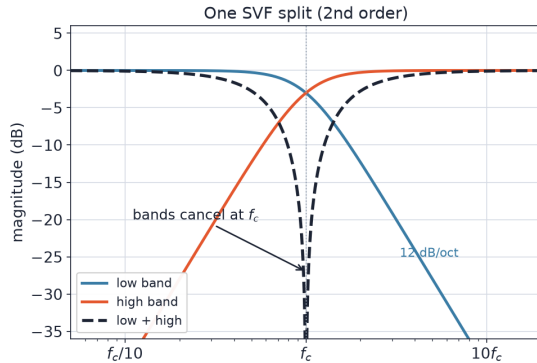
$$\text{low} + \text{high} = \text{original}$$

Recall the SVF filter computes its lowpass (v_2) and highpass (v_0) outputs *simultaneously*.

Tune it to f_c , take v_2 as the low band, v_0 as the high band. Done!

No! at f_c the two outputs have *equal magnitude* but are 180° *out of phase*. When we add the bands back together, they *cancel*; the recombined signal has a notch right at the crossover frequency.

The Linkwitz–Riley trick: filter twice



The Linkwitz–Riley trick: filter twice

The fix: run each band through the same filter *once more*. Each band's response gets **squared**.

Why does that help? The SVF's HP output is 180° out of phase with its LP output at *every* frequency.

So if we filter both bands a second time, the HP output is $360^\circ = 0^\circ$ out of phase! No cancellation. Also, we get a steeper cutoff.

This is the 4th-order **Linkwitz–Riley** crossover.

Exercise: multi-band distortion

Build a multiband distortion FX. We'll need three distinct state variable filters (each with their own state, but they can share coefficients)

1. **Split** into low/high at f_c with the first SVF (remember, the SVF produces low and high outputs from the same internal state).
2. **Filter** each low/high band a second time with the second and third SVFs (lowpass the low band, highpass the high band)
3. **Drive** each band, i.e. boost its gain, with the corresponding drive parameter.
4. **Shape** each band by applying the distortion function to each band.
5. **Sum** the two bands and apply the output gain.

Remember, filter state is per channel. So, for stereo audio, you need to keep separate state for the left and right channels of each filter.